ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

• AFFILIATED TO ANNA UNIVERSITY • CHENNAI



Harshini Raman

Data Architecture

Contact

+91-948616014

harshinidataarch@gmail.com

T2-Signa Apartments

About Me

Passionate Data architect with hands on expeirence in SQL,Python,Power Bi and Databricks

Skills

- Management Skills
- Creativity
- Communication Skill
- Negotiation
- Critical Thinking
- Leadership

Education

Bachelor of Engineering in Computer Science and Engineering 2026-2030

St Joseph's college of engineering

I am a budding Data Architect with a strong interest in designing, organizing, and optimizing data systems that support meaningful business insights and smarter decision-making. I am passionate about understanding how data moves across systems and how well-designed data architectures can turn complex information into reliable, scalable solutions.

AREAS OF INTEREST

Data Architecture · Data Modeling · Database Design · ETL/ELT · Data Engineering · Cloud Data Platforms · Data Analytics · Business Intelligence

My Vision

. To grow into a skilled Data Architect who can design robust data ecosystems, connect technology with business needs, and create data foundations that enable organizations to make confident, data-driven decisions.

Current Focus

Building practical knowledge through continuous learning, hands-on projects, and exploration of modern data technologies, with a focus on developing strong foundations in data architecture and engineering.

Career Goal

To contribute to innovative data-driven projects while continuously developing the technical and architectural skills required to design reliable, scalable, and future-ready data solutions.

Connect With Me

LinkedIn: linkedin.com/in/harshini-r-221a98422/

GitHub: github.com/Harshini478

Impressive Personality: Biographical Note

MATEI ZAHARIA

Creator of Apache Spark |Computer Scientist |Researcher |Technology Entrepreneur

---

Introduction

Matei Zaharia is a prominent computer scientist, researcher, professor, and technology entrepreneur best known for creating Apache Spark, a widely used open-source framework for large-scale data processing. His work has had a significant influence on distributed computing, cloud data systems, machine learning, and modern artificial intelligence infrastructure.

Early Academic Journey

Zaharia earned a Bachelor of Mathematics in Computer Science from the University of Waterloo and later completed his PhD in Computer Science at the University of California, Berkeley, in 2013. During his doctoral research, he focused on distributed systems and the challenges involved in processing very large amounts of data.

Creation of Apache Spark

In 2009, while pursuing his PhD at UC Berkeley, Zaharia and his research group began developing Apache Spark. The project was designed to make large-scale data processing faster, more flexible, and easier for researchers and engineers to use. Spark subsequently became a major open-source project and an important component of modern data engineering and analytics platforms.

Professional Career

Zaharia is a professor in the Electrical Engineering and Computer Sciences department at UC Berkeley and has continued to conduct research in computer systems and artificial intelligence. He also co-founded Databricks, a company focused on data and AI platforms, where he serves as Chief Technology Officer.

Research and Contributions

Beyond Apache Spark, Zaharia has contributed to a range of systems and research projects involving distributed computing, machine learning, databases, and artificial intelligence. His work includes contributions associated with projects such as Apache Mesos, MLflow, Delta Lake, and ColBERT. His research reflects a consistent interest in building systems that make complex computing tasks more efficient and accessible.

Personality and Professional Qualities

Zaharia's career demonstrates several qualities that are valuable in technology: intellectual curiosity, research-oriented thinking, persistence, innovation, and practical problem-solving. A notable feature of his work is the connection between academic research and real-world technology. He has repeatedly taken difficult technical problems and explored solutions that can be applied at large scale.

Recognition and Achievements

His contributions have received significant recognition in the computing community, including the ACM Doctoral Dissertation Award, the Presidential Early Career Award for Scientists and Engineers (PECASE), the SIGMOD Systems Award for Apache Spark, the Mark Weiser Award, and the ACM Prize in Computing in 2025.

Impact and Inspiration

Matei Zaharia's journey illustrates how academic research can develop into technology with broad practical impact. His work on Apache Spark helped shape the way organizations process and analyze large datasets, while his later work has extended into data platforms and artificial intelligence. For students and aspiring engineers, his career offers an example of how curiosity, strong technical foundations, research, and the ability to solve real problems can lead to meaningful contributions to the technology industry.

Why Matei Zaharia Is an Impressive Personality to Me?

He turned a research idea into a technology with real-world impact. Starting Apache Spark during his PhD shows how academic learning can be transformed into a practical solution. He combines strong technical knowledge with practical problem-solving. His work spans distributed systems, data engineering, cloud computing, databases, and artificial intelligence.He inspires me to build skills with purpose. His career shows that learning technical concepts is most valuable when those concepts are used to solve meaningful problems.He demonstrates continuous learning. Even after the success of Apache Spark, he continued working on technologies such as MLflow, Delta Lake, and Al systems.

His journey connects education, research, and industry. This is especially inspiring to me as an engineering student because it shows that a strong academic foundation can lead to significant professional contributions. He focuses on technology that can be useful at scale. His work demonstrates the importance of thinking beyond a small project and considering how a solution can help many people and organizations.

Key Takeaway

Matei Zaharia's professional journey can be summarized through four qualities: curiosity, innovation, problem-solving, and impact. His career demonstrates the potential of combining deep technical knowledge with practical thinking to create systems that influence an entire field.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps.

Covey's urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and "time stealers" such as waiting and travel can be turned into "time gifters."

The lectures trace procrastination to Parkinson's law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the "frog" (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying "yes" when you mean "no."

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely),

Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can’t buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

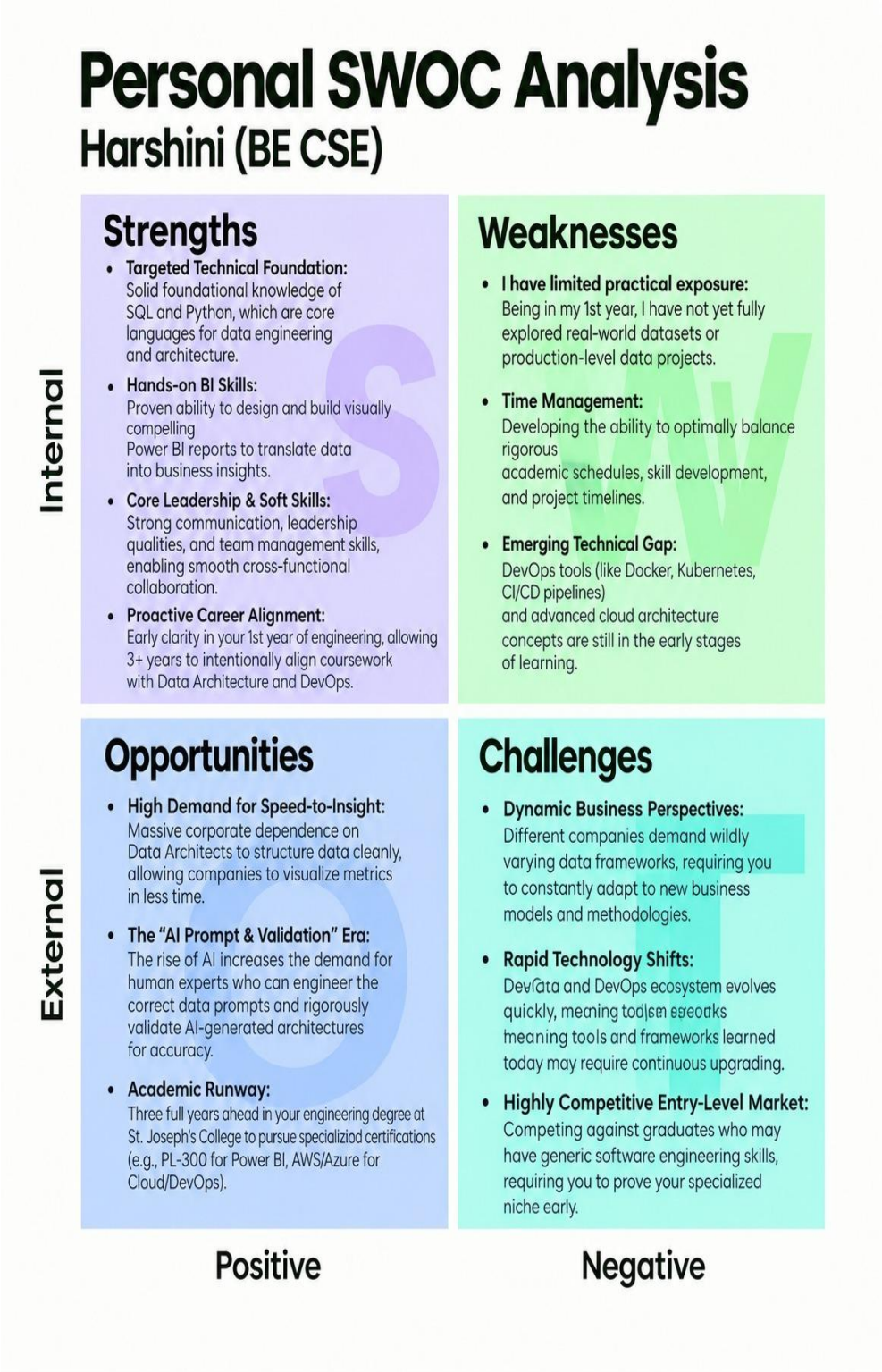
Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SWOC Analysis



SWOC CAREER ROADMAP

Target Role: Data Architect / Data Engineering & DevOps Professional
Harshini | 2026 | Engineering | 2026–2030 Batch

1. Title

- SWOC Career Roadmap
- Target Role: Data Architect / Data Engineering & DevOps Professional
- Harshini | 2026 | Engineering | 2026–2030 Batch
- Goal: Build strong data architecture and DevOps expertise and become a career-ready data professional.

2. Executive Summary & Career Vision

- **Vision:** Become a skilled Data Architect capable of designing scalable, reliable, and business-focused data solutions.
- **Objective:** Strengthen SQL, Python, Power BI, cloud, data architecture, and DevOps capabilities through academics, certifications, and practical projects.
- **Framework:** SWOC Analysis → Skill Development → Certifications → Projects → Industry Exposure → Career Readiness

3. SWOC Matrix at a Glance

STRENGTHS	WEAKNESSES
• Targeted technical foundation • Hands-on Power BI skills • Leadership & communication • Early career alignment	• Limited practical exposure • Time-management development • DevOps/cloud technical gap • Limited production experience
OPPORTUNITIES	**CHALLENGES**
• Growing demand for data architecture • AI prompt & validation • Three-year academic runway • Cloud & DevOps certifications	• Changing business requirements • Rapid technology shifts • Competitive entry-level market • Need for continuous upskilling

4. Strengths

- **Targeted Technical Foundation —** Strong foundational knowledge of SQL and Python provides an excellent base for data engineering and architecture.
- **Hands-on BI Skills —** Experience with Power BI helps transform data into meaningful visual insights and business reports.
- **Core Leadership & Soft Skills —** Strong communication, leadership, and team-management abilities support effective collaboration in technical teams.
- **Proactive Career Alignment —** Starting career preparation early gives several years to intentionally align coursework with Data Architecture and DevOps.

5. Weaknesses

- **Limited Practical Exposure —** As a first-year engineering student, I have not yet worked extensively with real-world datasets or production-level data projects.
- **Time Management —** I need to improve how I balance academic responsibilities, skill development, and project timelines.
- **Emerging Technical Gap —** Docker, Kubernetes, CI/CD pipelines, and advanced cloud architecture are still areas where I am developing my knowledge.
- **Industry Experience Gap —** I need more exposure to professional workflows, enterprise data systems, and real-world deployment environments.

6. Opportunities

- **High Demand for Speed-to-Insight —** Organizations increasingly need data professionals who can structure data efficiently and help businesses obtain insights quickly.
- **The AI Prompt & Validation Era —** AI growth creates opportunities for professionals who can design reliable data systems and validate AI-generated outputs.

- **Academic Runway —** Several years remaining in my engineering degree gives me time to build specialized skills before entering the job market.
- **Specialized Certifications —** Certifications such as PL-300, AWS, and Azure can strengthen my professional profile in BI, cloud, and DevOps.

7. Challenges

- **Dynamic Business Perspectives —** Different organizations use different data frameworks and business models, so I must develop the ability to adapt.
- **Rapid Technology Shifts —** Data and DevOps technologies evolve quickly, requiring continuous learning and regular skill upgrades.
- **Highly Competitive Entry-Level Market —** I will compete with graduates having broad software skills, so I need to demonstrate a clear specialization in data architecture.
- **Balancing Breadth and Depth —** Learning SQL, Python, BI, cloud, DevOps, and architecture together can be challenging, so I need a structured progression.

8. Action Plan: Closing the Gaps

Gap	Action Plan	Deadline
Limited practical exposure	Work with real datasets and complete 2 data projects	June 2027
Time management	Follow weekly study + project schedule	Start immediately
DevOps knowledge gap	Learn Docker, Kubernetes & CI/CD fundamentals	December 2027
Cloud knowledge gap	Build foundational AWS/Azure skills	June 2028
Credential gap	Complete PL-300 / relevant cloud certification	December 2028

9. Action Plan: Using Opportunities

- Use the academic runway: Align college projects and coursework with data engineering and architecture.
- Build a portfolio: Create SQL, Python, Power BI, cloud, and data-pipeline projects.
- Earn specialized credentials: Target PL-300, followed by AWS/Azure certifications.
- Explore AI + Data: Learn how AI can support data analysis while developing strong validation skills.
- Gain industry exposure: Pursue internships, hackathons, workshops, and industry projects from the second year onward.

10. 4-Year Milestones Matrix

Year	Primary Focus	Target Deliverable	Target Credential
Year 1	SQL, Python & Power BI	2 BI/data projects	Power BI/SQL certification
Year 2	Data Engineering & Cloud	Data pipeline + cloud project	PL-300
Year 3	DevOps & Architecture	Internship + deployed data project	AWS/Azure certification
Year 4	Advanced Architecture & Career	Major capstone + professional portfolio	Advanced cloud/data credential

11. Conclusion & Key Takeaways

- **Takeaway 1:** My foundation in SQL, Python, Power BI, communication, and leadership gives me a strong starting point for a data-focused career.
- **Takeaway 2:** My biggest priority is converting this foundation into practical experience, cloud/DevOps expertise, certifications, and industry exposure.
- **Verdict:** By using my four-year academic runway strategically, I can turn my current data and BI foundation into a strong specialization in Data Architecture and DevOps.

3-Minute Speaking Script

Slides 1–2

Good morning everyone. My presentation is about my SWOC Career Roadmap. My target career is in Data Architecture, with a focus on Data Engineering and DevOps. My goal is to develop the technical and professional skills required to design reliable and scalable data solutions. My vision is to become a skilled Data Architect who can connect technical data systems with real business requirements. I plan to achieve this through SWOC analysis, continuous skill development, certifications, projects, and industry exposure.

Slide 3

This is my SWOC matrix. My strengths include a good foundation in SQL and Python, hands-on Power BI skills, leadership and communication abilities, and early career alignment. My weaknesses are limited practical exposure, developing time-management skills, and gaps in DevOps and cloud architecture. At the same time, I have opportunities through the growing demand for data professionals, AI, certifications, and the academic runway available during my engineering degree. My main challenges are changing business requirements, rapid technology shifts, competition, and the need to balance different technical areas.

Slide 4

Coming to my strengths, my SQL and Python foundation gives me an important starting point for data engineering and architecture. My Power BI skills help me convert data into useful visual insights. My communication and leadership abilities will help me collaborate effectively with technical and business teams. Most importantly, I am thinking about my career early, which gives me enough time to deliberately build the skills required for my target role.

Slide 5

However, I also recognize my weaknesses. As a first-year student, I have limited exposure to real-world datasets and production systems. I also need to become better at balancing academics, projects, and skill development. Another important gap is my knowledge of Docker, Kubernetes, CI/CD, and advanced cloud architecture. Finally, I need professional exposure so that I understand how data systems work outside the academic environment.

Slides 6–7

These weaknesses also point toward my opportunities. Organizations increasingly need professionals who can structure data efficiently and deliver insights quickly. The growth of AI also creates opportunities for people who understand data quality and can validate AI-generated outputs. Since I still have several years of engineering ahead, I have a valuable academic runway to develop specialized skills. Certifications such as PL-300 and AWS or Azure credentials can further strengthen my profile. However, the industry is changing quickly, business requirements vary between organizations, and entry-level competition is high. I therefore need to remain adaptable and continuously upgrade my skills.

Slides 8–9

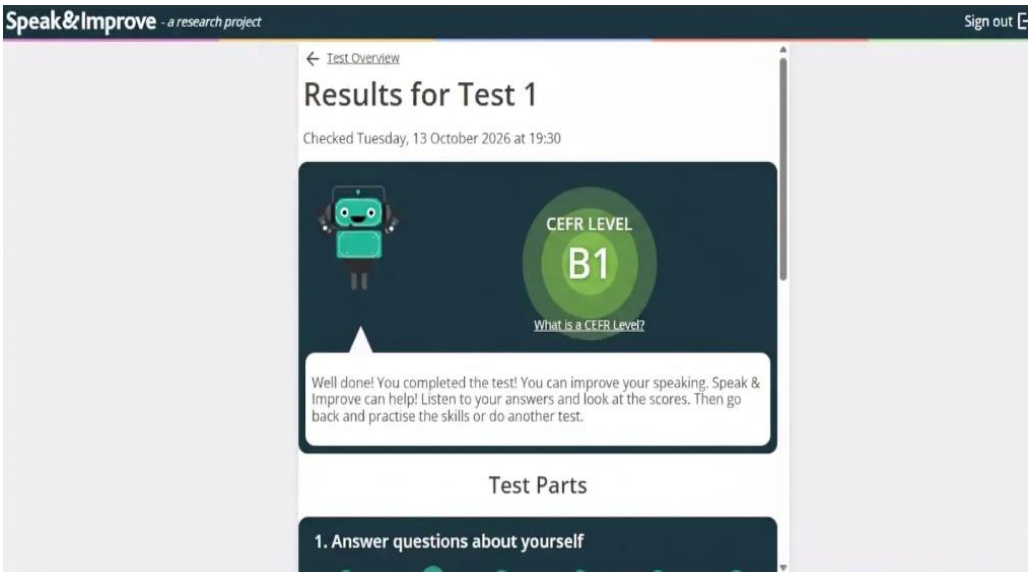
My action plan focuses on converting these gaps into strengths. I will gain practical experience through real datasets and projects, improve my time management through a structured weekly routine, and gradually learn Docker, Kubernetes, CI/CD, and cloud architecture. I also plan to use my academic years strategically by building a portfolio, pursuing certifications, exploring AI and data applications, and gaining internship experience. This approach will help me move from theoretical knowledge toward professional capability.

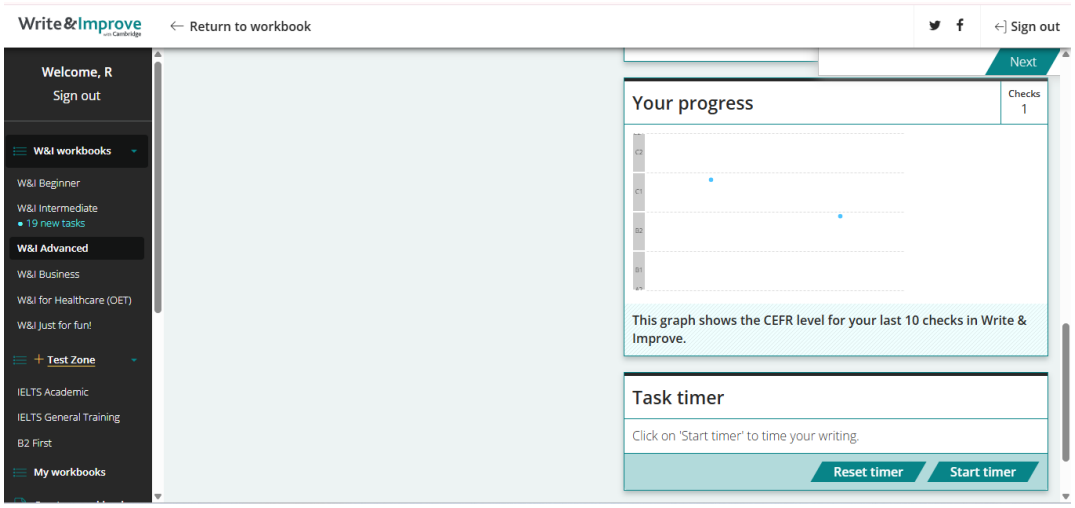
Slides 10–11

My four-year roadmap gives me a clear progression. In the first year, I will strengthen SQL, Python, and Power BI. In the second year, I will focus on data engineering and cloud and target PL-300. In the third year, I will develop DevOps and architecture skills and gain internship experience. In the final year, I will focus on advanced architecture, my major project, and career preparation. To conclude, my existing technical foundation is a strong starting point, but practical experience, cloud and DevOps knowledge, certifications, and industry exposure will determine my growth. My verdict is that if I use these four years strategically, I can build a strong specialization in Data Architecture and DevOps. Thank you.

Write and Improve / Speak and Improve

Speak and Improve: score sheet





Attending the Cambridge One Speak & Improve and Write & Improve modules has been an incredibly rewarding and transformative experience that has substantially enhanced my overall English language fluency and communicative competence.

Successfully achieving a CEFR B1 rank across both the speaking and reading/writing modules has provided me with an objective, internationally recognized baseline of my current abilities, giving me a clear and concrete understanding of where I stand as an independent English user. To validate this achievement, I have attached a screenshot proof of my official rank results below.

What makes these specific tools exceptionally valuable is their sophisticated, instant feedback system. Rather than simply receiving a static final score, I was presented with an interactive learning process where I could see exactly which grammatical structures required refinement and where my vocabulary could be expanded to sound more natural. In the speaking module, practicing with the automated platform forced me to focus heavily on my pronunciation, sentence pacing, and

conversational flow. This continuous oral practice drastically reduced the natural hesitation I usually experience when speaking, helping me transition from overthinking my words to delivering sentences more instinctively.

Similarly, the writing and reading components allowed me to experiment with diverse sentence patterns and expressions, showing me in real-time how to structure my thoughts more logically, improve paragraph coherence, and correct errors on the spot. Practicing in this supportive, pressure-free environment allowed me to treat every mistake as a direct learning opportunity, which built both my technical skills and my personal confidence. Ultimately, these Cambridge tests did not just evaluate my current proficiency; they served as an active, practical catalyst for my language development, leaving me highly motivated, confident, and well-equipped to start working toward the B2 level.

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is

more comfortable for long hours of detailed work.

3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully
10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Never share personal information such as your address, phone number or passwords with strangers online.
2. Always use strong passwords to protect your online accounts.
3. Avoid clicking on unknown links, suspicious messages and unwanted advertisements.
4. Use trusted websites while browsing, downloading files or making online payments.
5. Be careful while using social media and avoid accepting friend requests from unknown people.
6. Respect others online and never involve yourself in cyberbullying or spreading false information.
7. Keep your devices and antivirus software updated to protect them from viruses and cyberattacks.
8. Report cyberbullying, online fraud or suspicious activities to parents, teachers or the concerned authorities.

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Tamil Nadu is famous for its rich culture and traditional art forms. One of the most beautiful traditional art forms in my region is **Bharatanatyam**. It is a classical Indian dance form that originated in Tamil Nadu. It combines graceful movements, facial expressions, hand gestures and rhythmic footwork to express stories and emotions.

Bharatanatyam has been practised for many centuries and is an important part of Tamil culture. Dancers wear colourful traditional costumes, jewellery and anklets. The dance performances usually tell stories from Indian mythology and highlight moral values. It requires dedication, discipline and years of practice.

However, due to modernization and changing lifestyles, many traditional art forms are slowly losing their popularity among younger generations. Therefore, it is important to take steps to preserve them.

Modern initiatives can play a major role in protecting Bharatanatyam. Schools and colleges can organize cultural programmes, dance competitions and workshops to encourage students to learn this art. Social media platforms like YouTube and Instagram can be used to share performances and create awareness among people worldwide. Online dance classes can also help learners from different places access professional training.

The government and cultural organizations should provide scholarships, financial support and opportunities to talented artists. Traditional dance festivals and cultural events can also attract more people towards this art form.

In conclusion, Bharatanatyam is not just a dance but a valuable part of our cultural identity. By combining traditional practices with modern technology and initiatives, we can preserve this beautiful art form for future generations. It is our responsibility to respect, promote and protect our cultural heritage.

Question 28 (b) Formal letter of request to the Principal

Harshini Raman
BE CSE 2026-30

St.Joseph's College of Engineering

OMR,Chennai-130

2nd November,2026

The Principal,

St.Joseph's College of Engineering

OMR,Chennai-130

Subject: Request for Permission to Attend a Technical Workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am writing this letter to request your permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies.

The workshop will provide valuable knowledge about the latest developments in Artificial Intelligence, Machine Learning, Robotics and other emerging technologies. It will also help me

improve my technical skills, gain practical knowledge and understand the future opportunities in the field of technology.

I assure you that I will attend the workshop sincerely and complete any missed academic work. I kindly request you to grant me permission to participate in this workshop.

Thank you for your consideration.

Yours faithfully,
Harshini Raman

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment

Education plays an important role in shaping students' future. Assessment is an essential part of education because it helps measure students' knowledge and performance. Nowadays, Artificial Intelligence (AI) is being used in education for evaluating students. Both AI-based assessment and traditional teacher-based assessment have their own advantages and disadvantages.

AI-based assessment uses technology and computer programs to evaluate students' answers, conduct online tests and provide quick feedback. It saves time, reduces workload and gives instant results. It can also identify students' learning difficulties and suggest areas for improvement. However, AI may not always understand creative answers, emotions or individual student needs correctly.

Traditional teacher-based assessment involves teachers evaluating students through written exams, assignments, projects and classroom activities. Teachers can understand students' abilities, creativity and individual learning needs. They can provide personal guidance and encouragement. However, manual evaluation takes more time and may sometimes involve human bias.

When comparing both methods, AI-based assessment is faster and more efficient, while teacher-based assessment provides personal understanding and human judgement. AI can evaluate objective questions accurately, whereas teachers are better at assessing creativity, critical thinking and communication skills.

In conclusion, both AI-based and traditional teacher-based assessments are useful in education. The best approach is to combine both methods. AI can support teachers by reducing their workload, while teachers can provide personal guidance and fair evaluation. Together, they can improve the quality of education and create a better learning experience for students.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: Joe@gmail.com

Subject: First day in my college

Hi dear Friend!!

Hope you are doing well! I am very excited to share my experience of the orientation programme held at my college recently.

The programme was organized to welcome the new students and introduce them to college life. It started with a welcome speech by our Principal, who explained the college rules, facilities and academic activities. The teachers introduced themselves and gave useful guidance about our courses and career opportunities.

There were also cultural performances, fun games and interactive sessions. We were introduced to our departments, library, laboratories and sports facilities. The seniors were friendly and helped us feel comfortable in the new environment.

Overall, it was an enjoyable and informative programme. I made many new friends and learned a lot about my college. I really wish you had been there with me!

Take care and write back soon.

With warm regards,
Harshini

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must

2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning

2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have

overheated.

4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.

5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.

   (b) The team collected plenty of useful information from the field test.

2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.

   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.

3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.

   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.

4. (a) All electrical hardware and wiring must comply with local safety legislation.

   (b) The data scientist discovered several interesting insights after running the machine learning clustering.

5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.

   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.

   (b) The automated script runs every midnight to archive legacy server logs.

2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.

   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.

3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.

(b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.

4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately
   (b) instantly
3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

A Traditional Harvest Festival and Eco-Friendly Practices

Pongal is one of the most important traditional harvest festivals celebrated in Tamil Nadu. It is a festival of joy and thanksgiving, celebrated to thank the Sun God, farmers, cattle and nature for providing a good harvest. It is usually celebrated in January for four days.

People clean and decorate their houses with colourful kolams. They prepare a special dish called Pongal using freshly harvested rice, milk and jaggery. Families wear traditional clothes, visit relatives and share happiness. Cattle are decorated and worshipped as they play an important role in agriculture.

Nowadays, people are encouraged to celebrate Pongal in an eco-friendly way. Natural materials such as banana leaves, clay pots and flowers can be used for decorations. We should avoid plastic decorations and disposable items. Organic products and locally grown food should be preferred. Food waste should be reduced, and leftover food can be shared with others. We can also plant trees and keep our surroundings clean.

Eco-friendly celebrations help protect the environment and preserve our traditions. Pongal teaches us to respect farmers, nature and the importance of agriculture.

In conclusion, Pongal is not just a harvest festival but also a symbol of gratitude, unity and cultural heritage. We should celebrate it responsibly by following eco-friendly practices and passing these values to future generations.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

The Temple Car Festival is one of the most beautiful traditional festivals celebrated in Tamil Nadu. It is an important cultural and religious event that brings people together. During this festival, the beautifully decorated temple car carrying the deity is pulled through the streets by devotees. People gather with great devotion and celebrate with happiness and enthusiasm.

Faith plays an important role in this festival. Devotees offer prayers, sing devotional songs and seek blessings. The streets are decorated with colourful lights, flowers and traditional kolams.

Food is another special attraction of the festival. People prepare traditional dishes and distribute free food to devotees. Many volunteers arrange drinking water, snacks and meals for visitors. This shows the spirit of sharing and unity.

However, large gatherings may create waste and pollution. We can make the celebration cleaner by avoiding plastic plates, cups and covers. Instead, we can use banana leaves and reusable plates. Dustbins should be placed in public areas, and people should avoid throwing waste on the streets. Volunteers can organize cleaning activities after the festival. We should also save water and maintain cleanliness around the temple.

In conclusion, the Temple Car Festival represents faith, tradition and unity. By following eco-friendly practices, we can protect our environment while enjoying our cultural celebrations. A clean festival is a happy festival.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

Folk art and handicraft fairs are an important part of Indian culture and tradition. They provide a platform for talented artists and craftsmen to display and sell their handmade products. These fairs help preserve our cultural heritage and promote traditional skills.

In Tamil Nadu, many traditional art forms such as pottery, wood carving, palm-leaf crafts, Tanjore paintings and traditional jewellery are popular. These fairs attract many visitors and provide opportunities for local artists to earn money. People can learn about different cultures, traditions and artistic skills through these exhibitions.

Handicraft fairs also encourage eco-friendly practices. Most handmade products are created using natural materials such as clay, bamboo, jute, wood and palm leaves. Using these products helps reduce plastic pollution. Organizers can avoid plastic bags and use paper or cloth bags instead. They should also provide separate dustbins for waste collection and maintain cleanliness in the exhibition area.

Modern technology can help promote traditional handicrafts through online exhibitions, websites and social media. This allows artists to reach customers from different parts of the world.

In conclusion, folk art and handicraft fairs play an important role in protecting our cultural heritage and supporting local artists. By adopting eco-friendly practices, we can preserve our traditions and create a greener future for the next generation.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30

a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Harshini Raman

I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

2 November 2026

To

The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for Permission and Lab Access After College Hours to Prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to develop a simple AI-based solution to solve a real campus problem using Python and Machine Learning. We need approximately three weeks to design, train and test the model. Unfortunately, our personal computer systems are not powerful enough for this purpose.

We require access to the laboratory between 4:30 p.m. and 7:30 p.m. on weekdays, along with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection.

We assure you that we will follow all laboratory rules, maintain the equipment carefully and leave the laboratory clean and switched off after use. We are also willing to work under the supervision of a faculty member, if required.

We would be very grateful if you could grant us permission to use the laboratory facilities. Thank you for your kind consideration.

Yours obediently,

Harshini Raman

Question 2: Letter to the Plant Manager (industrial plant visit)

Harshini Raman

I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

2 November 2026

To

The Plant Manager
[Company Name]
[Company Address]

Subject: Request for Permission to Arrange an Industrial Visit

Respected Sir/Madam,

I am writing on behalf of the students of the first-year Computer Science and Engineering department to request permission to visit your industry as part of our academic learning programme.

The main objective of this industrial visit is to gain practical knowledge about industrial operations, modern technologies, manufacturing processes and workplace management. It will help students understand how theoretical concepts are applied in real-world industries.

We are planning to organize the visit on [date] with approximately 40 students, accompanied by faculty members. We kindly request you to provide us with an opportunity to visit your plant and interact with your technical staff.

We assure you that all students will follow the safety instructions and industrial rules during the visit. We will maintain discipline and cooperate with your staff throughout the programme.

We would be grateful if you could grant us permission for this industrial visit. Thank you for your time and consideration.

Yours faithfully,

Harshini Raman
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Harshini Raman

I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

2 November 2026

To

The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for Permission to Access the Machine Shop for an Electric Go-Kart Project

Respected Sir/Madam,

I am writing on behalf of my project team to request permission to use the Mechanical Engineering Department's machine shop for developing an electric go-kart as part of our technical project.

The main objective of this project is to design and develop an electric go-kart that demonstrates practical engineering skills and promotes eco-friendly transportation. We require access to the machine shop to perform basic fabrication, cutting, drilling and assembly work for the vehicle frame and other components.

We kindly request permission to use the machine shop during the available working hours under the supervision of a faculty member or laboratory technician. We would also appreciate guidance regarding the available tools and equipment required for our project.

We assure you that all team members will strictly follow workshop safety rules, wear the necessary protective equipment and handle the machines responsibly. We will also maintain cleanliness and ensure that all equipment is returned properly after use.

We would be grateful if you could grant us permission to access the machine shop and support us in completing this project successfully.

Thank you for your kind consideration.

Yours faithfully,

Harshini Raman
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Harshini Raman

I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

2 November 2026

To

The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for Sponsorship and Duty Leave to Participate in a National-Level Technical Event

Respected Sir/Madam,

I am writing on behalf of my project team to request financial sponsorship and duty leave to participate in an upcoming National-Level Technical Event.

Our team has been selected to participate in a National AI Hackathon, which provides an excellent opportunity to showcase our technical skills, creativity and innovative ideas. Participating in this event will help us gain practical knowledge, improve our problem-solving abilities and interact with students from various institutions.

We kindly request the college management to provide financial support towards registration fees, travel expenses and other necessary arrangements. We also request duty leave for the event dates, [mention dates], so that we can participate without affecting our attendance.

We assure you that we will represent our college with dedication and maintain good discipline throughout the event. We will also complete any missed academic work after returning.

We would be grateful if you could consider our request and provide the necessary sponsorship and duty leave.

Thank you for your kind support and consideration.

Yours faithfully,

Harshini Raman
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
(On behalf of the project team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Transportation plays an important role in our daily lives. With the development of technology, electric vehicles (EVs) are becoming popular alternatives to conventional internal combustion engine (ICE) vehicles. Both types of vehicles are used for transportation, but they differ in their working principles, fuel sources, cost and environmental impact.

Electric vehicles operate using electric motors powered by rechargeable batteries. They do not produce exhaust emissions while driving. They are quieter, require less maintenance and are energy-efficient. However, they need charging stations and may take more time to recharge compared to refuelling conventional vehicles.

On the other hand, conventional vehicles use petrol or diesel engines to produce power. They can be refuelled quickly at petrol stations and are widely available. However, they release harmful gases such as carbon dioxide and other pollutants, which contribute to air pollution and global warming. They also require regular engine maintenance.

Electric vehicles generally have lower running and maintenance costs, while conventional vehicles may have higher fuel and maintenance expenses. However, the initial purchase price of electric vehicles can be higher, depending on the model.

Both types of vehicles have advantages and disadvantages. Electric vehicles are a better option for reducing local air pollution, while conventional vehicles currently offer wider refuelling infrastructure and convenience for long-distance travel.

In conclusion, electric vehicles represent an important step towards sustainable transportation. With improvements in battery technology and charging infrastructure, EVs can become more practical and affordable. Choosing cleaner transportation methods will help protect the environment and create a greener future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes

immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight

tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: teacher@gmail.com

Respected Ma'am,

I hope you are doing well. I am writing this email to share my experience of my first semester at college. I really miss my school days and your valuable guidance.

My first semester at St. Joseph's College of Engineering has been an exciting and memorable experience. Initially, I felt nervous about the new environment, but gradually I became comfortable with my classmates and teachers. The college provided many opportunities to improve my technical knowledge and communication skills.

I attended lectures, laboratory sessions, workshops and various activities. I also made new friends and learned to manage my time and responsibilities. Some subjects were challenging, but with regular practice and support from my teachers, I was able to understand them better.

Your guidance during my school days helped me build confidence and face these new challenges. I am truly thankful for your encouragement and support.

Please convey my regards to all my school teachers. I hope to meet you soon and share more of my college experiences.

Yours sincerely,
Harshini

Question 2: Email to a high school science teacher about the first term

To: teacher@gmail.com

Respected Ma'am,

I hope you are doing well. I am happy to write to you and share my experiences from my first semester at St. Joseph's College of Engineering. I often remember your encouragement and the valuable lessons you taught me during my school days.

College life has been a completely new experience for me. I have learned many interesting concepts in Computer Science and Engineering, especially programming and computer applications. Practical

laboratory sessions have helped me understand how theoretical knowledge is applied in real-life situations.

Apart from academics, I have made many new friends and participated in department activities. I also attended technical sessions that introduced me to Artificial Intelligence and modern technologies. These experiences have improved my confidence, communication skills and ability to work in a team.

Initially, adjusting to the college timetable and managing assignments was a little difficult. However, I am slowly learning to balance my studies and extracurricular activities. For my upcoming examinations, I have planned to revise my subjects regularly, practise programming and prepare well for practical examinations.

I am truly grateful for the strong foundation and motivation you gave me during school. Your teaching continues to inspire me to learn new things and work towards my goals.

Please convey my regards to all my school teachers. I hope to meet you soon!

Yours sincerely,
Harshini Raman

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation

Campus Bus Crowd Predictor

1. Problem Definition

Campus buses can become overcrowded during peak hours, while some buses may have empty seats at other times. Students may not know how crowded a bus will be before they board it. This can cause discomfort, delays, and poor use of available buses.

The project aims to predict the expected crowd level of campus buses using previous travel data and current information such as time, route, day, and special campus events.

2. Proposed Technical Architecture

The system will collect bus-related data such as passenger count, route, date, time, and bus number.

High-level workflow:

Data Collection →ETL Pipeline →Database →Prediction Model → API → Power BI Dashboard

- Data can be collected through a simple student/driver form or an existing campus transport system.
- An ETL pipeline cleans and prepares the collected data.
- The processed data is stored in a database.
- A prediction algorithm analyzes previous patterns and predicts the expected crowd level.
- An API can provide the prediction to a web or mobile interface.
- Power BI can display crowd trends, peak hours, busy routes, and predicted crowd levels.

The system can classify buses as Low, Medium, or High Crowd instead of predicting an exact number of passengers.

3. Target Users & Social/Economic Impact

Target Users

- Students
- Faculty and staff
- Transport coordinators
- Campus administration

Impact

Students can choose less crowded buses and plan their travel better. Transport coordinators can understand peak travel periods and make better decisions about bus allocation.

The system can also help reduce unnecessary bus trips, improve resource usage, and reduce waiting time and overcrowding.

4. Feasibility & Constraints

Tools

- Python for data processing and prediction
- SQL/database for storing data
- Power BI for dashboards
- ETL tools or Python for data pipelines
- REST API for connecting the prediction system

Cloud platform such as AWS or Azure for future deployment

- Git/GitHub for version control and DevOps practices

Constraints

- Accurate prediction depends on having enough historical data.
- Initial data may need to be collected manually.

Student privacy must be protected.

- Real-time GPS or passenger sensors may not be available.

Budget & Deployment Scope

A basic prototype can be developed with existing software tools and sample or manually collected campus data, keeping the initial cost very low. The first version can cover one campus and a limited number of bus routes.

In the future, GPS, automated passenger counting, and real-time data can be added to improve the system.

Project Pitching

Good morning everyone.
Our project is called Campus Bus Crowd Predictor.
The problem we are trying to solve is overcrowding in campus buses. During peak hours, some buses become very crowded, while other buses may still have empty seats. Students often do not know the crowd level before choosing a bus. This can lead to discomfort, waiting time, and inefficient use of buses.
Our solution is a data-based system that predicts the expected crowd level of a campus bus.
We will collect data such as the bus route, time, day, passenger count, and special events. This data will go through an ETL pipeline, where it will be cleaned and prepared. It will then be stored in a database.
A prediction model will study past patterns and classify the expected crowd as low, medium, or high.
The results can be shown through a Power BI dashboard, where students and transport coordinators can see crowd trends, busy routes, and peak hours.
The project is practical because we can start with manually collected data and later add GPS or automated passenger-counting systems.
Our goal is to use data to make campus transport more predictable, comfortable, and efficient.
Thank you.

Personal Pitching

Hi, I'm Harshini, a first-year CSE engineering student. My key strengths are good communication and leadership skills. I have worked on creating Power BI reports using real-life data and have also worked with ETL pipelines. I'm interested in building my career in Data Architecture and DevOps. I want to use my current skills, gain more technical knowledge, and work on real-world projects to create useful solutions
Good morning everyone. I'm Harshini, and I'm a first-year CSE engineering student. I'm happy to have this opportunity to introduce myself.
I chose Computer Science because I'm interested in technology and I enjoy learning how data and software can be used to solve real-world problems.
My key strengths are good communication and leadership qualities. I enjoy working with people, sharing ideas, and taking responsibility when working in a team. I believe these skills will help me both in my studies and in my future career.
I have also worked on some practical projects using real-life data. One of my key experiences was creating Power BI reports. Through this, I learned how to work with data and present it in a clear and useful way. I have also worked with ETL pipelines, which helped me understand how data can be collected, processed, and moved for further use.
These experiences have increased my interest in the data field. My career goal is to build a career in Data Architecture and DevOps. I want to strengthen my technical skills, work on more real-world projects, and gain practical experience in these areas.
As I continue my engineering journey, I want to keep learning, improve my technical knowledge, and use my communication and leadership skills to contribute effectively to my team.
Thank you for giving me this opportunity to introduce myself

Linkedin Profile - Harshini Raman

Aspiring Data Architect | Computer Science and Engineering Student | Interested in Data, AI & Emerging Technologies

About

I am a Computer Science and Engineering student at St. Joseph's College of Engineering, passionate about learning how technology can solve real-world problems.

I am particularly interested in programming, computer applications, Artificial Intelligence and emerging technologies. I enjoy exploring new technical concepts, improving my problem-solving abilities and gaining practical knowledge through laboratory learning and technical activities.

My career goal is to become a **Data Architect**, designing and developing efficient, reliable and scalable data systems. I am interested in understanding how data is collected, stored, organised, processed and used to support better decision-making.

I believe in continuous learning, teamwork and adapting to new technologies. I look forward to building my technical knowledge and gaining hands-on experience through projects and learning opportunities.

Focus Areas: Data Architecture • Programming • Artificial Intelligence • Database Management • Data Management • Emerging Technologies • Problem Solving

Featured Projects

As I continue developing my technical skills, I aim to build projects related to:

- **Database Management System:** Designing and organising structured data.
- **Student Data Management:** Developing an application to store and manage student information.
- **Data Analysis Project:** Exploring datasets and identifying useful insights.
- **AI-Based Applications:** Understanding how Artificial Intelligence can solve practical problems.

Education

Bachelor of Engineering – Computer Science and Engineering (B.E. CSE)
St. Joseph's College of Engineering, OMR, Chennai.

Interests

- Data Architecture and Database Management
- Artificial Intelligence and Machine Learning
- Programming and Software Development
- Computer Applications
- Emerging Technologies
- Technical Sessions and Department Activities
- Practical Learning and Teamwork

Skills

Area	Focus
Technical Interests	Programming, AI, Computer Applications
Data Technologies	Database Management, Data Architecture
Professional Skills	Communication, Teamwork
Learning Strengths	Adaptability, Practical Learning, Problem Solving

Career Roadmap

Stage	Learning Focus	Planned Goal
Year 1	Programming and computer science fundamentals	Build a strong technical foundation
Year 2	SQL, DBMS and data modelling	Develop database-based projects
Year 3	Data engineering, cloud fundamentals and ETL	Build practical data projects
Year 4	Data architecture and big data technologies	Develop a data architecture capstone project

Connect with me on:

Name: Harshini Raman
LinkedIn: https://www.linkedin.com/in/harshini-r-221a98422/

I'm Harshini Raman

CSE Student | Aspiring Data Architect | AI & Technology Enthusiast

Welcome to my GitHub profile!

About Me

- I am a Computer Science and Engineering student at St. Joseph's College of Engineering.
- Interested in programming, Artificial Intelligence and modern technologies.
- Aspiring to become a Data Architect.
- Exploring database management, data organisation and computer applications.
- Interested in teamwork, practical learning and technical activities.
- Always curious to learn new technologies and improve my skills.

Areas of Interest

- Data Architecture
- Database Management Systems (DBMS)
- Artificial Intelligence and Machine Learning
- Programming and Software Development
- Data Engineering
- Cloud Technologies
- Data Analysis

Technical Learning Goals

Domain	Learning Focus
Programming	Programming fundamentals
Databases	SQL and DBMS
Data Architecture	Data modelling and database design
Data Engineering	ETL and data pipelines
Artificial Intelligence	AI and ML fundamentals
Cloud	Cloud-based data platforms

Projects

I am working towards developing practical projects in areas such as:

- Student Data Management System
- Database Design and Management
- Data Analysis Projects
- AI-Based Applications

Project repositories will be added as I develop and publish my work.

My Career Goal

My long-term goal is to become a **Data Architect** and gain expertise in designing, managing and developing scalable data systems.

My Learning Journey

- Strengthen computer science and programming fundamentals.
- Learn SQL and database management.
- Explore data modelling and data engineering.
- Gain practical experience through projects.
- Explore cloud technologies and modern data platforms.
- Work towards becoming a skilled Data Architect.

Connect With Me

- **GitHub:** https://github.com/Harshini478
